

Software Quality Assurance & Software Maintenance

Question Bank (1, 2, 3, 4 Marks)

PART A: 1 Mark Questions

- 1 Define Quality in software engineering.
- 2 What is Software Quality Assurance (SQA)?
- 3 What is a Software Review?
- 4 What is a Formal Technical Review (FTR)?
- 5 Define Software Reliability.
- 6 What is ISO 9000?
- 7 What does CMM stand for?
- 8 What is Six Sigma?
- 9 What is an SQA Plan?
- 10 Define Software Maintenance.
- 11 Name any one type of Software Maintenance.
- 12 What is Re-engineering?
- 13 What is Reverse Engineering?
- 14 What is Forward Engineering?

PART B: 2 Marks Questions

- 1 State any two objectives of Software Quality Assurance.
- 2 Differentiate between Verification and Validation.
- 3 Write the purpose of Formal Technical Review (FTR).
- 4 List the characteristics of Software Reliability.
- 5 Write two benefits of ISO 9000 standards.
- 6 What are the five maturity levels of CMM?
- 7 Write two principles of Six Sigma in software engineering.
- 8 List the components of an SQA Plan.
- 9 Differentiate between Corrective and Adaptive Maintenance.
- 10 Differentiate between Reverse Engineering and Re-engineering.

PART C: 3 Marks Questions

- 1 Explain the concept of Software Quality.
- 2 Explain Software Quality Assurance activities.
- 3 Describe the process of Formal Technical Review.
- 4 Explain Software Reliability and its importance.
- 5 Explain ISO 9000 standards in software engineering.

- 6 Explain the Capability Maturity Model (CMM).
- 7 Describe Six Sigma methodology for software engineering.
- 8 Explain the types of Software Maintenance.
- 9 Explain Reverse Engineering with example.
- 10 Explain Re-engineering and Forward Engineering.

PART D: 4 Marks Questions

- 1 Explain Software Quality Assurance in detail and describe the contents of an SQA Plan.
- 2 Explain Software Reviews and Formal Technical Review (FTR) with advantages and disadvantages.
- 3 Explain Software Reliability and Quality Standards (ISO 9000, CMM, Six Sigma) in detail.
- 4 Explain the types of Software Maintenance with suitable examples.
- 5 Explain Re-engineering, Reverse Engineering, and Forward Engineering in detail with comparison.